

§ T1 Endogenous depth of reasoning

1.1 Research question(s) addressed in this topic

本课题探讨了博弈中策略思维的深度是如何内生决定的。研究重点在于如何通过成本收益分析来解释推理深度，即参与者如何根据激励（收益）和认知能力来权衡进一步思考的价值，以及这种深度如何与对对手的信念相互作用。该模型试图解决传统 level-k 模型中推理步数是外生给定的这一缺陷。

This topic explores how the depth of strategic thinking in games is endogenously determined. The research focuses on explaining reasoning depth through a cost-benefit analysis, where players weigh the value of extra introspection against cognitive costs based on payoffs and abilities, and how this depth interacts with beliefs about opponents. This framework attempts to address the limitation in traditional level-k models where the number of reasoning steps is exogenously given.

1.2 Summarize the design and purpose of the set of experiments in this topic

实验旨在验证推理步数是否随激励和信念变化。实验采用修改后的 11-20 博弈，两名被试匹配后在 11 到 20 之间选一个整数，并获得该数值的代币。如果选择比对手小 1 的数字，可获得额外奖励 \$x\$。为了操纵激励，奖励 \$x\$ 分为 20 或 80 代币两档；若数字相同，则各得 10 代币。研究者通过学科背景（理工 vs 文科）或特定的认知测试（高分 vs 低分）将被试分类并贴上标签，以此操纵他们对对手能力的信念。实验还包含“替代处理” C，让被试与两个特定标签的人对局中产生的真实结果进行博弈，用于测试高阶信念的影响。

The experiments aim to test whether reasoning steps vary with incentives and beliefs. The experiment uses a modified 11-20 game where paired subjects choose an integer between 11 and 20 and receive that amount in tokens. A bonus of \$x\$ is awarded for choosing exactly one less than the opponent. To manipulate incentives, the bonus \$x\$ was set at either 20 or 80 tokens, while a tie yielded 10 tokens. Beliefs about opponents were manipulated by labeling subjects based on their majors (Math/Science vs. Humanities) or test performance (High vs. Low scores). A "replacement treatment" C was also included, where subjects played against a number chosen in a game between two other players of a specific label to test higher-order beliefs.

1.3 Summarize the main experimental findings in this topic

Subjects increase their reasoning depth when stakes are higher and when they believe opponents are more sophisticated. Belief manipulation via labels produces shifts in choices consistent with theory, and overall, the results support the idea that reasoning depth is endogenously determined by a cost-benefit evaluation. Therefore, it is problematic to treat observed actions as a direct measure of a fixed cognitive ability.

被试在高奖励下或认为对手更复杂时，推理深度增加。通过标签操纵对手信念导致选择变化方向与理论一致，总体结果支持推理深度由成本收益权衡内生决定的观点，因此把观察到的行为直接当成“固定认知能力”的做法会有问题。

1.4 Critical thoughts on the experimental design and the authors' interpretation

First, the experiment emphasizes only one-shot initial responses. Suppressing feedback helps cleanly observe first-move reasoning, but limits external validity because real strategic interaction is usually repeated with feedback, learning, and adjustment. Second, belief manipulation through labels may not be strong enough: weaker subjects may not understand or update beliefs about the opponent, so observed behavior changes could reflect label effects rather than actual belief shifts.

首先，实验只强调一次性的初始反应。抑制反馈有助于干净观察首次决策的推理，但限制了外部效度，因为现实中的策略互动通常是重复的，会有反馈、学习和调整。其次，通过标签操纵信念的力度可能不够强：推理能力较弱的被试可能无法理解或更新对手信念，因此观察到的行为变化可能反映的是标签效应，而非真实的信念更新

Arch

§ T2 Network Experiment

1.1 Describe the research question(s) addressed in this topic.

本主题研究网络结构如何影响个体在战略互补与战略替代环境下的决策行为。它对比了完全信息与不完全信息两种信息结构，探讨这些环境如何共同塑造个体的策略选择。此外，研究重点在于当博弈存在多个均衡解时，网络特征如何引导参与者最终选择哪一种特定的均衡。

This topic investigates how network structure influences individual decision-making under strategic complements and strategic substitutes. It compares complete and incomplete information structures to explore how these environments jointly shape strategic choices. Furthermore, the research focuses on how network features guide participants to select a specific equilibrium when multiple equilibria exist.

1.2 Summarize the design and purpose of the set of experiments in this topic.

实验设计旨在探究网络结构如何影响个体在战略互补与战略替代环境下的决策。在受控的实验室中，参与者被分配到具有不同平均度、连通性和聚类系数的随机网络中。实验对比了两种信息结构：完全信息下参与者可见全图，而不完全信息下参与者仅知晓自身度数与网络概率分布。每位参与者需在活跃与不活跃之间进行二元选择，其收益取决于邻居的行动。实验通过多轮重复博弈，目的是验证网络特征如何引导群体在存在多个均衡时进行均衡选择。

The experimental design aims to investigate how network structure influences individual decision-making in environments of strategic complements and strategic substitutes. In a controlled laboratory setting, participants are assigned to random networks with varying average degrees, connectivity, and clustering coefficients. The experiment compares two information structures: under complete information, participants can see the entire graph, while under incomplete information, they only know their own degree and the network's probability distribution. Each participant makes a binary choice between active and inactive, with payoffs depending on their neighbors' actions. Through multiple rounds of repeated play, the purpose is to verify how network features guide group equilibrium selection when multiple equilibria exist.

1.3 Summarize the main experimental findings in this topic.

当均衡唯一时，被试行为与理论预测非常接近，尤其在不完全信息下也依然如此；而当存在多个均衡时，均衡选择会系统地受到 degree、connectivity 和 clustering 的影响。具体来说，实验 1 表明，在完全信息下，战略替代环境中被试往往选择较安全但略低于最优效率的均衡，而在战略互补环境中，只要存在清晰的高收益协调结构，就往往能够实现有效率均衡；在不完全信息下，行为总体仍符合 threshold-style prediction，但网络不确定性会削弱向高效率均衡的协调。实验 2 和实验 3 进一步表明，较高的 degree 和 connectivity 会提高活跃率，clustering 也有促进协调的作用，虽然其效应相对较弱。

Behavior is very close to the theoretical prediction whenever equilibrium is unique, and this remains true even under incomplete information; when multiple equilibria exist, equilibrium selection is systematically shaped by degree, connectivity, and clustering. More specifically, under complete information, subjects in strategic substitutes tend to select a safer equilibrium that is slightly less than fully efficient, whereas in strategic complements they often coordinate on the efficient equilibrium when there is a clear high-payoff coordination structure. Under incomplete information, behavior still broadly follows threshold-style predictions, but network uncertainty weakens successful coordination on efficient outcomes. Further experiments show that higher degree and greater connectivity increase activity rates, while clustering also tends to support coordination, though more modestly.

1.4 Do you have any critical thoughts on the experimental design and the authors' interpretation of their results? If so, specify them.

我认为该实验设计虽然逻辑清晰，但在外部效度和效应分离上存在局限。首先，实验将行动简化为“活跃”或“不活跃”的二元选择，虽然这便于识别均衡，但忽略了现实中互动强度（如努力程度）的连续性，限制了结论在复杂场景下的适用性。其次，由于实验采用多轮重复并提供反馈，后期的行为可能混合了学习效应和路径依赖，而不仅是对静态不完全信息的反应，这使得单纯由网络结构驱动的效应难以被彻底分离。

I believe that while the experimental design is logically clear, it has limitations regarding external validity and the isolation of effects. First, simplifying actions into a binary choice of "active" or "inactive" helps identify equilibria but overlooks the continuous intensity of real-world interactions (such as effort levels), limiting the results' applicability to complex settings. Second, since the experiment uses repeated rounds with feedback, later behaviors likely mix in learning effects and path dependence rather than being purely static responses to incomplete information, making it difficult to isolate the effects driven solely by network structure.

Arch

§ T3 Ambiguity aversion

1.1 Describe the research question(s) addressed in this topic.

本课题的核心研究问题是探讨人们在面对“未知概率”（模糊性）时的决策行为，以及这种行为与处理“复合风险”（多阶段概率）之间的内在联系。具体来说，研究者想弄清楚为什么大多数人表现出模糊规避（即偏好已知概率），以及这种偏好是否源于人们无法按照概率法则将复杂的多阶段抽奖简化为单阶段抽奖（即违反“复合抽奖简化”原则，ROCL/RCLA）。此外，进一步探讨了不同形式的“部分模糊”（如区间、不相交和两点分布）如何影响这种决策倾向。

The core research question of this topic is to explore how people make decisions under "unknown probabilities" (ambiguity) and the internal link between this behavior and the processing of "compound risks" (multi-stage probabilities). Specifically, researchers aim to understand why most people exhibit ambiguity aversion (preferring known probabilities) and whether this preference stems from a failure to reduce complex multi-stage lotteries into single-stage ones according to probability laws (violating the "Reduction of Compound Lottery" axiom, ROCL/RCLA). Additionally, Chew et al. further investigate how different forms of "partial ambiguity"—such as interval, disjoint, and two-point distributions—affect these decision-making tendencies.

1.2 Summarize the design and purpose of the set of experiments in this topic.

实验设计主要分为经典的埃尔斯伯格悖论扩展和 Chew 等人的部分模糊实验。在 Halevy 的实验中，受试者面对四个球罐：已知比例罐、完全未知罐、均匀分布的复合风险罐和两点分布的复合风险罐。实验通过 BDM 机制或价格表法让受试者写下每个赌注的“保留价格”（确定性等值），以此衡量他们的偏好。Chew 等人的实验则更细致，使用了 100 张红黑卡片，通过改变卡片组成的限制条件（如红卡在 $\$50-n\$$ 到 $\$50+n\$$ 之间）来模拟不同的模糊程度，并设置对应的客观复合风险组。其目的是为了测试不同理论模型（如递归预期效用模型 REU 和递归秩相关效用模型 RRDU）的预测准确性，并验证模糊态度与复合风险态度之间的关联。

The experimental design is primarily divided into extensions of the classic Ellsberg paradox and the partial ambiguity experiments by Chew et al. In Halevy's setup, subjects faced four urns: a known-risk urn, a fully unknown urn, a uniformly distributed compound-risk urn, and a two-point distributed compound-risk urn. The experiments elicited "reservation prices" (certainty equivalents) for each bet using the BDM mechanism or price-lists to measure preferences. Chew et al.'s experiment was more granular, using decks of 100 red and black cards and simulating various levels of ambiguity by restricting possible compositions (e.g., red cards ranging from $\$50-n\$$ to $\$50+n\$$), while setting up corresponding objective compound-risk groups. The purpose was to test the predictive accuracy of different theoretical models, such as Recursive Expected Utility (REU) and Recursive Rank-Dependent Utility (RRDU), and to verify the link between attitudes toward ambiguity and compound risk.

1.3 Summarize the main experimental findings in this topic.

实验的主要发现是模糊中性与复合抽奖简化（ROCL）之间存在极强的相关性：那些能够正确简化复合概率的人，通常在模糊面前也是中性的。相反，大多数表现出模糊规避的人也表现出对复合风险的规避。具体到部分模糊，研究发现随着可能组合数量的增加或分布跨度的扩大，受试者的确定性等值（CE）普遍呈下降趋势。此外，研究证明了人群是异质的，大约 70% 的受试者可以归类为递归模型（REU 或 RRDU）所描述的类型，这说明模糊规避本质上与处理多阶段不确定性的方式有关。The primary finding is a tight association between ambiguity neutrality and the Reduction of Compound Lottery (ROCL): individuals who correctly reduce compound probabilities are typically ambiguity neutral. Conversely, most people who exhibit ambiguity aversion also show aversion to compound risks. Regarding partial ambiguity, the studies found that certainty equivalents (CE) generally decrease as the number of possible compositions or the spread of the distribution increases. Furthermore, the results demonstrate population heterogeneity, with about 70% of subjects being classifiable under recursive models (REU or RRDU), suggesting that ambiguity aversion is fundamentally linked to how individuals process multi-stage uncertainty.

1.4 Do you have any critical thoughts on the experimental design and the authors' interpretation of their results? If so, specify them.

首先，BDM 机制在认知上非常复杂，受试者可能并未真正理解其激励相容的逻辑。如果受试者感到困惑，他们报出的“保留价格”可能只是随机噪音，或者倾向于选择某个中间值（如 1 美元或 10 美元）作为焦点，而无法代表真实的确定性等值（CE），这削弱了数据的准确性。其次，实验样本仅限于大学生，这种高度同质化的群体存在严重的代表性问题。这些受试者通常具有更强的逻辑和概率处理能力，使得研究结论很难推广到缺乏此类背景的大众群体中。

First, the BDM mechanism is cognitively demanding, and subjects might not fully grasp its incentive-compatible logic. If subjects are confused, their reported "reservation prices" may reflect random noise or a tendency to pick focal points (e.g., \$1 or \$10) rather than true certainty equivalents (CE), which undermines the accuracy of the data. Second, the experimental sample is restricted to university students, leading to significant representativeness issues. These participants typically possess higher logical and probabilistic skills, making it difficult to generalize the findings to the broader public who lack such backgrounds.

1.1 Describe the research question(s) addressed in this topic.

研究问题关注消费者在面对不完全显著的销售税时如何反应。具体来说，实验考察消费者是否低估税收影响，这种低反应在不同个体之间有何差异，以及提高税率是否会改变消费者的注意力和反应程度

The research questions address how consumers respond to sales taxes that are not fully salient. Specifically, the experiments investigate whether consumers underreact to taxes, how this underreaction varies across individuals, and whether increasing the tax rate affects consumers' attention and responsiveness.

1.2 Summarize the design and purpose of the set of experiments in this topic.

实验样本来自美国 45 个有销售税的州。设计包含被试间和被试内变化。被试被随机分到无税、标准税或三倍税三组。在 Module 1 中，各组分别面对 0 倍、1 倍和 3 倍本地销售税，而在 Module 2 中，所有组都面对无税环境。每个模块中，被试对 20 个家庭用品报告他们愿意支付的最高 tag price，采用 BDM 支付机制，并设置理解题和实验末 survey。该设计的目的在于识别平均低反应、个体异质性，以及税率增大时注意力是否提升

Respondents used from the 45 US states with positive sales taxes. The design combines between-subject and within-subject variation. Subjects were randomly assigned to a no-tax, standard-tax, or triple-tax arm. In module 1 they faced zero, standard, or triple local sales taxes, while in module 2 all groups faced no tax. In each module, subjects stated the highest tag price they would pay for 20 household products, with BDM incentives. The purpose was to identify average underreaction, heterogeneity across individuals, and whether attention rises when taxes become larger.

1.3 Summarize the main experimental findings in this topic.

当消费者面对不显著的销售税时，他们往往会低估其影响。这种低反应在不同个体之间存在差异，并且随着税率上升而增强，表明注意力会受到税率大小的影响。消费者通常对税率有一定了解，因此这些错误并非源于无知，而是由于注意力或处理能力有限。个体之间存在明显异质性：有的消费者几乎完全考虑税收，有的部分考虑，有的几乎不考虑。考虑这种异质性对于估算税收效率损失具有重要意义。

Consumers tend to underreact to sales taxes when they are not salient. This underreaction varies across individuals and increases when taxes are higher, indicating that attention is influenced by the magnitude of the tax. Consumers generally know the tax rates, so mistakes are not due to ignorance but limited attention or processing. There is notable heterogeneity: some consumers fully account for taxes, some partially, and some barely at all. Accounting for this variation has substantial implications for estimating the efficiency costs of taxation.

1.4 Do you have any critical thoughts on the experimental design and the authors' interpretation of their results? If so, specify them.

One limitation of the experimental design is that it only considers relatively small household products, so results may not generalize to high-stakes or essential purchases where attention and salience might differ.

The online setting may also reduce validity compared to in-store behavior, also, some respondents have trouble understanding the experiment, which may dispute the result.

实验设计的一个局限是只涉及相对小额的家庭用品，因此结果可能无法推广到高价值或必需品的购买场景，这类场景中注意力和税收显著性可能不同。

在线实验环境也可能降低有效性，还有被试不理解实验，可能影响最终的结果。

§ T5 Stereotypes

1.1 Describe the research question(s) addressed in this topic.

This chapter studies how people form and use stereotypes when judging groups. It asks whether individuals overweight representative types, how this affects their beliefs, and whether stereotypes systematically exaggerate differences between groups.

本章节研究人们在评估群体时如何形成和使用刻板印象。它探讨个体是否会过度重视具有代表性的类型，这如何影响他们的信念，以及刻板印象是否系统性地夸大了群体间差异。

1.2 Summarize the design and purpose of the set of experiments in this topic.

The purpose was to show that stereotypes stem from "representative" traits that distinguish one group from another. The design used a 15-second exposure, filler tasks, and an assessment. In the T-shirt experiment, the girl group remained constant while only the reference (boy) group was changed. Results showed people overweight a color if it is unique to a group, even if it is not frequent. This proves stereotypes are context-dependent and shaped by comparison.

其目的在于表明刻板印象源自于区分一个群体与另一个群体的“代表性”特征。实验设计采用了 15 秒的展示时间、填充任务以及评估环节。在 T 恤实验中，女孩组保持不变，而仅改变参照（男孩）组。结果表明，如果某种颜色是某个群体所独有的，即便其出现频率不高，人们也会对其过度重视。这证明了刻板印象是依情境而定的，并且是通过比较形成的。

1.3 Summarize the main experimental findings in this topic.

Participants overweight representative types when making judgments, producing biased beliefs that exaggerate differences between groups. Stereotypes are context dependent: the same target group is evaluated differently depending on the reference group. This supports the representativeness heuristic as a mechanism for stereotype formation.

被试在判断时过度重视代表性类型，导致信念偏差，夸大群体间差异。刻板印象受上下文影响：同一目标群体在不同参考群体下会得到不同评价。这支持代表性启发式是刻板印象形成的机制。

1.4 Do you have any critical thoughts on the experimental design and the authors' interpretation of their results? If so, specify them.

实验设计的一个核心问题在于 15 秒的展示时间过于短暂。在这么短的时间内，受试者根本无法准确记住所有信息。因此，实验结果很可能只是反映了在注意力有限的情况下的瞬时记忆扭曲。这种基于短暂视觉残留的认知偏差，或许并不能等同于现实生活中那种根深蒂固、带有社会偏见的刻板印象。

A major flaw in the design is that the 15-second exposure is too short. Within such a brief window, participants cannot accurately memorize the details. Consequently, the results likely reflect memory distortion caused by limited attention rather than actual belief formation. These instant cognitive glitches may not truly represent the deep-seated social stereotypes found in the real world.

§ T6 Trust and betrayal

1.1 Describe the research question(s) addressed in this topic.

本主题主要探讨信任是否为人类经济行为中的一种“基本原始驱动力”(economic primitive)，以及哪些因素会影响信任的产生与丧失。具体研究问题包括：在排除声誉、合同和惩罚等机制后，信任是否依然存在；人们在面对自然风险与人为带来的“社会风险”(尤其是背叛)时，其风险承受意愿是否存在差异；以及受托人能否通过“信息设计”作为一种信号设备，来主动提升信誉并促成信任行为。This topic primarily explores whether trust is an "economic primitive" in human behavior and what factors increase or decrease its likelihood. Specific research questions include: can trust persist when mechanisms like reputation, contracts, and punishment are eliminated; is there a significant difference in risk-taking when the source of uncertainty is another person (betrayal) rather than nature; and can trustees use "information design" as a signaling device to boost trusting acts.

1.2 Summarize the design and purpose of the set of experiments in this topic.

实验 1 的核心操作是“投资博弈”。A 房间的被试拥有 10 美元初始资金，并可以决定发送其中的一部分给 B 房间的匿名同伴。这笔钱在传输过程中会翻三倍。B 房间的人收到钱后，再决定保留多少或返还多少。这一设计的目的是通过双盲和单次匿名环境，观察在没有声誉激励或惩罚威胁时，人类是否依然存在信任和互惠这种基本行为。

实验 2 采用了“最小可接受概率”(MAP)法来量化风险意愿。参与者面临两个选择，即稳拿 10 点收益或选择一个可能得到 15 点也可能只有 8 点的风险选项。他们需要写下愿意接受风险选项的最低概率门槛。该实验在三种情境下进行：一是对手决定结果，二是自然随机决定结果但对手获益，三是纯粹的个人风险任务。这种对比是为了剥离风险偏好和社会偏好，从而精准测量“背叛厌恶”。

实验 3 设计了一个“信息设计”框架。受托人(玩家 A)首先通过选择 p 值来决定产生高回报状态的概率。接着他们还要设计一套信号结构，比如确定在不同状态下球的颜色分布。委托人(玩家 B)在观察到信号结构和随机抽取的球色后，最后决定是否投资。这种安排是为了研究受托人能否通过提前承诺信息透明度来缓解信任困境。Experiment 1 utilizes an "investment game" framework. Participants in Room A receive \$10 and can send any portion to an anonymous partner in Room B. The sent amount is tripled by the experimenter before it reaches the partner. The partner then decides how much of the tripled total to return and how much to keep. This design aims to observe whether trust and reciprocity exist as basic human elements by eliminating reputation effects and punishment threats.

Experiment 2 employs the "Minimum Acceptable Probability" (MAP) method to quantify risk acceptance. Participants choose between a guaranteed or a risky option. They record the lowest success probability they would accept to choose the risky option. This is conducted in three settings: where an agent determines the outcome, where nature determines it with an agent benefiting, and where nature determines it alone. The purpose is to isolate "betrayal aversion" from social and risk preferences. Experiment 3 involves an "information design" setting. The trustee (Player A) first decides the probability p of a high-payoff state. They then design an information structure such as determining the distribution of colored balls for each state. After observing the structure and a signal, the trustor (Player B) makes the final investment decision. This examines if trustees can use information commitment as a signaling device to foster trust.

1.3 Summarize the main experimental findings in this topic.

实验结果证实信任确实是一种基本行为，即便在匿名单次博弈中人们仍会表现出信任与互惠。跨国研究发现，“背叛厌恶”是普遍存在的，即人们在面对人带来的风险时要求的成功概率（MAP）明显高于面对自然风险时。此外，引入信息设计能显著提升信誉和信任水平，受托人通常会利用高透明度的信号来展示其诚信，或通过模糊信号来掩盖较低的信誉。社会历史信息的提供也能增强人们对社会规范的内化，从而提高回馈比例。Findings confirm that trust is a behavioral primitive, as people exhibit trust and reciprocity even in anonymous, one-shot interactions. Cross-country evidence shows that "betrayal aversion" is robust, meaning individuals require a significantly higher MAP when another person determines the outcome compared to nature. Furthermore, information design significantly boosts trustworthiness and trust; trustees tend to signal high trustworthiness via high informativeness and obscure low trustworthiness with low informativeness. Providing social history also enhances the internalization of social norms, leading to higher payback rates.

1.4 Do you have any critical thoughts on the experimental design and the authors' interpretation of their results? If so, specify them.

质疑点在于实验 1 中“信任是基本原始驱动力”这一结论的外部有效性。实验中所使用的 10 美元资金是由研究者直接发放的出场费，而非被试通过劳动换取的报酬。经济学中著名的“温室资金”效应表明，人们在处理这种非劳动所得的意外之财时通常会表现得更加慷慨且愿意冒险。如果这笔初始资金是参与者经过高强度劳动辛苦赚来的，发送金额和信任水平很可能会大幅下降。因此，实验中观察到的高度信任可能只是在特定实验室激励机制下的产物，而在涉及真实个人财富或血汗钱的社会互动中，信任作为“基本原始驱动力”的作用可能远没有论文中体现的那么显著。

A critique challenges the external validity of the conclusion in Experiment 1 that trust is a "behavioral primitive." The \$10 used in the sessions was provided as a show-up fee by the experimenters rather than earned through the subjects' own labor. The well-known "house money effect" in economics suggests that individuals tend to be more generous and risk-prone when dealing with such windfall gains. If the initial endowment had been earned through hard labor, the willingness to send money and the resulting trust levels would likely decrease significantly. Consequently, the high degree of trust observed may only be a byproduct of specific lab-based incentive structures, and the role of trust as a "primitive" might be far less prominent in real-world interactions involving hard-earned personal wealth.

§ T7 Lying

1.1 Describe the research question(s) addressed in this topic.

本课题主要研究人们为何有时即便说谎能获得更多利益也会选择诚实。研究者旨在探讨说谎的内在成本，通过分析谎言在收益、结果和可能性这三个维度上的“大小”如何影响行为。此外，该研究还关注社会身份如何驱动人们为了维持诚实形象而牺牲金钱收益。

This topic mainly investigates why people sometimes choose to be honest even when lying would yield higher material payoffs. The researchers aim to explore the intrinsic costs of lying by analyzing how the "size" of a lie—measured across payoff, outcome, and likelihood dimensions—affects behavior. Furthermore, it examines how social identity drives individuals to sacrifice monetary gains to maintain an honest image.

1.2 Summarize the design and purpose of the set of experiments in this topic.

实验通过对比“已观察”和“不可观察”两种环境，系统探究说谎的内在成本。在已观察游戏中，参与者需在电脑上点击十个方块之一以显示数字、立陶宛单词或随机分配收益的数字，随后在纸上向实验者汇报结果。这一设计旨在通过“单词组”和“混合数字组”剥离结果距离与金钱收益对撒谎的影响。而在不可观察游戏中，参与者从信封中抽取纸条并放回，由于实验者无法核实真相，这为观察社会身份效应提供了空间。此外，研究还设置了“低概率组”将最高收益设为 1% 可能性，或增加到 100 种状态，以此测试先验概率和状态分布如何改变人们对诚实形象的维护策略。

The experimental design compares "observed" and "non-observed" settings to systematically investigate the intrinsic costs of lying. In the observed games, participants click one of ten boxes on a computer to reveal numbers, Lithuanian words, or mixed-payoff numbers, then report the outcome on paper. This setup, specifically the "Words" and "Numbers Mixed" treatments, aims to disentangle the distance of outcomes from monetary gains. In contrast, the non-observed games involve drawing a paper slip from an envelope and replacing it, creating a space where the experimenter cannot verify the truth, thus allowing for the study of social identity effects. Additionally, the research includes a "Low Probability" treatment where the top payoff has only a 1% chance, and a "100-States" treatment, to test how prior probabilities and the number of states alter the strategies for maintaining an honest image.

1.3 Summarize the main experimental findings in this topic.

研究发现虽然人们普遍会为了利益说谎，但其行为受到社会身份关注的强烈调节。在已观察组中，大多数撒谎者会直接汇报最高收益，但在不可观察组中，部分谎言（即虚报但未达最高额度）的比例显著提高，反映出人们倾向于牺牲部分收益来维持“并非骗子”的形象。数据还揭示了明显的撒谎阈值：当观测到高收益（如 9 或 10）时人们极少撒谎，而观测到低收益时则频繁虚报。此外，先验分布的影响十分关键，当最高收益出现的可能性降低时，人们为了避免显得像是在作弊，会显著增加部分谎言的汇报频率，导致撒谎的分布进一步扩大。

The findings indicate that while people generally lie for profit, their behavior is strongly regulated by social identity concerns. In the observed groups, most liars report the maximum payoff, but in non-observed settings, the frequency of partial lies—reporting more than the truth but less than the maximum—increases significantly as individuals sacrifice potential gains to maintain a non-liar persona. The data also reveals a clear threshold for lying: people rarely lie when they observe high-payoff outcomes (like 9 or 10) but frequently misreport when they observe low payoffs. Furthermore, prior distribution plays a crucial role; as the likelihood of the highest outcome decreases, participants significantly increase the frequency of partial lies to avoid appearing dishonest, thereby expanding the range of misreported values.

1.4 Do you have any critical thoughts on the experimental design and the authors' interpretation of their results? If so, specify them.

该模型假设说谎的内在成本在所有被试间均质，但实际上，个体的道德偏好或对被观察的敏感度各不相同，这会导致行为异质性，而模型未能捕捉到。其次，将不可观察游戏中较多的部分谎言直接归因于社会身份效应可能存在解释上的混淆。例如，被试可能感知到在不可观察环境中被抓到的风险较低，因此更容易撒谎，这抵消了维护诚实身份的效应。The model assumes that the intrinsic cost of lying is homogeneous across all subjects; however, in reality, individual moral preferences or sensitivity to being observed vary, leading to behavioral heterogeneity that the model fails to capture. Furthermore, directly attributing the higher frequency of partial lies in non-observed games to social identity effects may involve an interpretative confound. For instance, participants might perceive a lower risk of being caught in non-observed environments and thus be more inclined to lie, which offsets the effect of maintaining an honest identity.

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